

OLIVIA BROUILLETTE

763-999-0768 ◊ ob1@nmsu.edu

EDUCATION

Doctor of Philosophy in Astronomy
New Mexico State University, Las Cruces, NM

August 2024 - Expected May 2029

Bachelor of Science in Mathematics and Physics
Iowa State University, Ames, IA

August 2019 - December 2023

COMPUTING SKILLS

Programming Languages	Python, Fortran, MATLAB
Operating Softwares	Linux, MacOS
Softwares	Pencil code, Athena code, Systems toolkit, ArcGIS, Agisoft Metashape, Ansys,SAOIamgeDS9

RESEARCH EXPERIENCE

New Mexico State University
Current Research Projects

August 2024 - Present

- Compared seven hydrodynamics codes applied to planet formation with the Planet Formation in the Southwest collaboration
- Parameterize planetesimal formation via streaming instability using various parameters
 - Investigating dispersion for the streaming instability.
 - Simulating streaming instability based on calculated growth rate.

Iowa State University

· *Advisor: Jacob B. Simon*

September 2022 - May 2024

- Worked on computational astrophysics in planetesimal formation
 - Scan parameter space for numerical hydrodynamical simulations of streaming instability

· *Advisor: Halil Ceylan*

May 2021 - December 2021

- Worked with small Unmanned Aircraft Systems to inspect transportation infrastructure
 - Inspected airport runways, aprons, and taxiways using pre-made drone operations.
 - Generated 3D spatial data via Agisoft Metashape and ArcGIS Pro.

RESEARCH PUBLICATION

A Comparative Study of the Streaming Instability: Unstratified Models with Marginally Coupled Grains (2026), S.A. Baronett, W. Lyra, A. Hossam, **O. Brouillette**, Daniel Carrera, Victoria I. De Cun, Linn E. J. Eriksson, Mario Flock, Pinghui Huang, Leonardo Krapp, Geoffroy Lesur, Rixin Li, Shengtai Li, Jeonghoon Lim, Sijme-Jan Paardekooper, David G. Rea, Debanjan Sengupta, Jacob B. Simon, Prakruti Sudarshan, Orkan M. Umurhan, Chao-Chin Yang, Andrew N. Youdin, submitted to Astrophysical Journal, in review, Mar. 2026, doi:10.48550/arXiv.2603.04558.

The Streaming Instability in 3D: Conditions for Strong Clumping, J. Lim, J.B. Simon, R. Li, **O. Brouillette**, David G. Rea, Wladimir Lyra, Astrophysical Journal, Sept. 2025, doi: 10.48550/arXiv.2509.18270

Small Unmanned Aircraft System for Pavement Inspection: Task 4—Execute the Field Demonstration Plan and Analyze the Collected Data, Md. A. A. Sourav, H. Ceylan, C. Brooks, D. Peshkin, S. Kim, R. Dobson, C. Cook, M. Mahedi, **O. Brouillette**, Federal Aviation Administration ,Nov. 2022,
doi: <https://doi.org/10.21949/1524511>

TEACHING & OUTREACH

New Mexico State University

- *Astronomy*

- Teaching Assistant, ASTRO 305, The Search for Life in the Universe Spring 2025
- Teaching Assistant, ASTRO 1115, The Planets Fall 2024,2025

- *Outreach*

- Joint graduate outreach coordinator for the NMSU astronomy department August 2025-Present
 - 23 hours done over the course of the 2024-2025 academic year
 - 32 hours done of the course of the 2025-2026 academic year

Iowa State University

- Teaching Assistant, CE 2710, Engineering Foundations of Statics Fall 2020-Fall 2024
 - Head Teaching assistant Fall 2022-Fall 2024
- Teaching Assistant, PHYS 0500, Preparation for Introductory Physics Fall 2023